Monte Carlo Option Pricing

From Risk-Neutral Valuation to Trading Desk Practice

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Abstract

I build a complete Monte Carlo option pricing engine on 200,000 paths, covering European calls, barrier options, and Greeks-based P&L attribution. The engine is validated against Black-Scholes closed-form values (absolute error < 0.03). A delta hedging simulation under real-world drift confirms that the Black-Scholes price equals the cost of replication regardless of the stock's expected return. The Broadie-Glasserman-Kou correction reduces discrete monitoring bias on barrier options from \$0.23 to \$0.05. Finally, Student-t return simulation demonstrates that fat tails alone generate the volatility smile observed in markets. The full implementation is available as a companion Jupyter notebook.

1 Introduction

1.1 What is an option?

A call option gives the holder the right, but not the obligation, to buy an underlying stock at a predetermined strike price K at a future maturity date T. The holder pays a premium upfront and receives the payoff at maturity:

$$\text{Payoff} = \max(S_T - K,\ 0) = (S_T - K)^+. \tag{1}$$

If the stock ends above the strike ($S_T > K$), the holder exercises and profits $S_T - K$. Otherwise, the option expires worthless. A put option is the symmetric contract: the right to sell at K, with payoff $(K - S_T)^+$.

1.2 The pricing problem

Pricing an option means determining its fair value today, before the outcome is known. The key insight of modern financial theory [1] is that the fair price is not the expected profit of holding the option. Instead, it is the **cost of replicating** the option payoff by dynamically trading the underlying stock. This replication argument leads to the fundamental pricing equation:

$$C = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+], \tag{2}$$

where r is the risk-free rate and $\mathbb{Q}$ is the risk-neutral measure—a probability distribution under which the stock's expected return equals r regardless of its actual expected return μ. The reason this works is explained in Section 2.4.

2 The GBM Model

2.1 Stochastic differential equation

Stock prices are modelled as geometric Brownian motion (GBM):

$$dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t, \tag{3}$$

where μ is the drift (expected return per year), σ is the volatility (annualised standard deviation of log-returns), and W_t is a standard Brownian motion. The multiplicative form $S_t\, dW_t$ ensures prices stay positive.

2.2 Why classical calculus fails

To find the distribution of S_T, a natural idea is to take the logarithm: if $f(S) = \ln S$, then classical calculus gives $df = dS/S = \mu\, dt + \sigma\, dW_t$, implying $\ln(S_T/S_0) = \mu T + \sigma W_T$. But this is wrong.

The error comes from a property of Brownian motion: its increments satisfy $(dW_t)^2 = dt$ (not zero as in classical calculus). This means the second-order term in the Taylor expansion of $f(S_t)$ does not vanish.

2.3 Itô's lemma

For any twice-differentiable function $f(S_t)$ of a GBM process, Itô's lemma states:

$$df = f'(S)\,dS + \tfrac{1}{2}\,f''(S)\,\sigma^2 S^2\,dt. \tag{4}$$

The extra term $\frac{1}{2}f''\sigma^2 S^2\,dt$ has no analogue in classical calculus. Applying this to $f(S) = \ln S$ with $f' = 1/S$ and $f'' = -1/S^2$:

$$d\ln S_t = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma\,dW_t. \tag{5}$$

The $-\sigma^2/2$ is the **Itô correction**. Integrating from 0 to T:

$$S_T = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,Z\right], \tag{6}$$

where $Z \sim \mathcal{N}(0,1)$. This is an exact solution—no discretisation error.

Remark. Without $-\sigma^2/2$, $\mathbb{E}[S_T]$ would equal $S_0 e^{(\mu+\sigma^2/2)T}$—biased high due to Jensen's inequality applied to the convex exponential function. With the correction, $\mathbb{E}[S_T] = S_0 e^{\mu T}$ as required. In the simulation, the mean of 200,000 terminal prices matches $S_0 e^{rT}$ to four decimal places.

2.4 Risk-neutral pricing

A trader who sells a call option and hedges by holding $\Delta = \partial C/\partial S$ shares eliminates first-order exposure to the stock. The profit from this hedged portfolio depends on the stock's random movements but—crucially—**not on its expected return** μ. The μ terms cancel because the stock position gains exactly what the option position loses (and vice versa) to first order.

Since the hedged portfolio is risk-free, it must earn r by no-arbitrage. This means the option price satisfies a PDE (the Black-Scholes PDE) in which μ does not appear. Equivalently, the price can be computed as an expectation under a modified measure $\mathbb{Q}$ where the drift is replaced by r:

$$S_T^{\mathbb{Q}} = S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,Z\right]. \tag{7}$$

This is what "risk-neutral pricing" means: the option price is independent of μ. The delta hedging simulation in Section 5 confirms this empirically.

3 Black-Scholes Formula

The expectation in Eq. (2) with $S_T^{\mathbb{Q}}$ from Eq. (7) can be computed analytically. The payoff is positive when $S_T > K$, which occurs when $Z > z^*$ where $z^* = \frac{\ln(K/S_0)-(r-\sigma^2/2)T}{\sigma\sqrt{T}}$. Evaluating the Gaussian integral by completing the square:

$$C = S_0\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2), \tag{8}$$

$$d_1 = \frac{\ln(S_0/K) + (r+\sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \tag{9}$$

Interpretation of d_2: $\Phi(d_2)$ is the risk-neutral probability that the option expires in the money ($S_T > K$). The term $Ke^{-rT}\Phi(d_2)$ is the present value of the strike payment, weighted by the probability of exercise.

Interpretation of d_1: $\Phi(d_1)$ is the probability of exercise under the stock-numéraire measure. It also equals Δ, the number of shares to hold for hedging.

The put price follows from **put-call parity**: $P = C - S_0 + Ke^{-rT}$.

With our parameters ($S_0 = 100$, $K = 105$, $T = 1$, $r = 5\%$, $\sigma = 20\%$): BS Call = 8.0214, BS Put = 7.9004.

4 Monte Carlo Estimation

The MC estimator for Eq. (2) is:

$$\hat{C}_N = e^{-rT} \cdot \frac{1}{N}\sum_{i=1}^{N}(S_T^{(i)} - K)^+, \tag{10}$$

where each $S_T^{(i)}$ is drawn from Eq. (7). The standard error SE $= \hat{\sigma}/\sqrt{N}$ gives the convergence rate $O(1/\sqrt{N})$: halving the error requires 4× as many paths.

With $N = 200{,}000$: the MC call price is 8.0398 vs BS exact 8.0214 (error 0.018). Both call and put fall within the 95% CI.

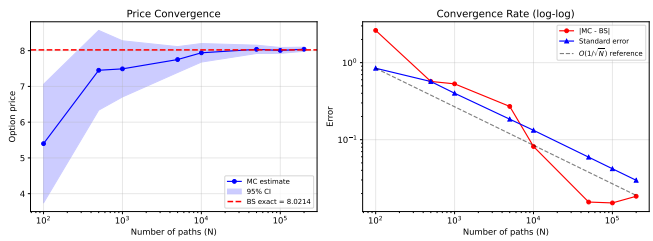

Figure 1: MC convergence. **Left:** price estimate with 95% CI narrows around the BS exact value (red) as N increases. **Right:** log-log error plot confirms the theoretical $O(1/\sqrt{N})$ convergence rate (slope $= -0.500$).

5 Delta Hedging Simulation

5.1 The replication mechanism

A trader who sells a call at the BS price C_0 can replicate the payoff:

1. **At** $t = 0$: receive premium C_0, buy $\Delta_0 = \Phi(d_1) \approx 0.54$ shares at cost $\Delta_0 S_0$. Remaining cash: $C_0 - \Delta_0 S_0$ (the trader borrows).
2. **At each rebalancing date**: recompute Δ_t from BS. Buy or sell $(\Delta_t - \Delta_{t-1})$ shares. Cash earns r.
3. **At maturity**: sell all shares. Pay $(S_T - K)^+$ to the holder.

5.2 Simulation under real-world drift

I simulate 50,000 paths under the **real-world measure** with drift $\mu = 8\%$—deliberately different from $r = 5\%$.

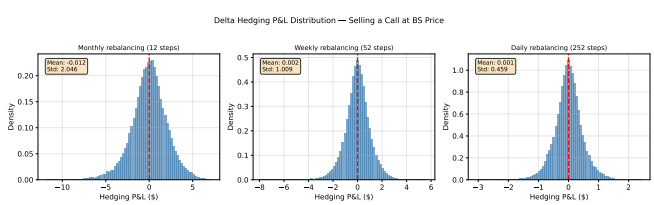

Figure 2: Hedging P&L at three rebalancing frequencies. All three distributions are centred on zero (red dashed line), confirming that the BS premium exactly covers the hedging cost regardless of the real-world drift $\mu = 8\%$. The spread narrows from monthly to daily: more frequent rebalancing reduces the hedging error.

Three findings (Figure 2, Table 1):

(i) Mean P&L ≈ 0 at all frequencies. This confirms risk-neutral pricing: the real-world drift μ does not affect the hedging cost.

(ii) P&L standard deviation shrinks with frequency. Monthly: 25.5% of C_0; daily: 5.7%. In the continuous limit, it would be zero.

(iii) The residual variance is the "gamma P&L." Between rebalancing dates, the delta changes and the cost of catching up is proportional to $\Gamma(\Delta S)^2$.

Table 1: Hedging P&L summary ($C_0 = 8.02$).

Frequency	Mean	Std	Std/C_0
Monthly (12)	≈ 0	2.05	25.5%
Weekly (52)	≈ 0	1.01	12.6%
Daily (252)	≈ 0	0.46	5.7%

6 Greeks and P&L Attribution

6.1 The Greeks

The Greeks are partial derivatives of the BS call price $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$:

Table 2: Black-Scholes Greeks for a European call.

Greek	Formula	Meaning
$\Delta = \Phi(d_1)$	$\partial C/\partial S$	Shares to hold
$\Gamma = \dfrac{\phi(d_1)}{S\sigma\sqrt{T}}$	$\partial^2 C/\partial S^2$	How fast Δ changes
$\mathcal{V} = S\phi(d_1)\sqrt{T}$	$\partial C/\partial \sigma$	Vol sensitivity
Θ	$\partial C/\partial t$	Time decay per day

where ϕ is the standard normal PDF.

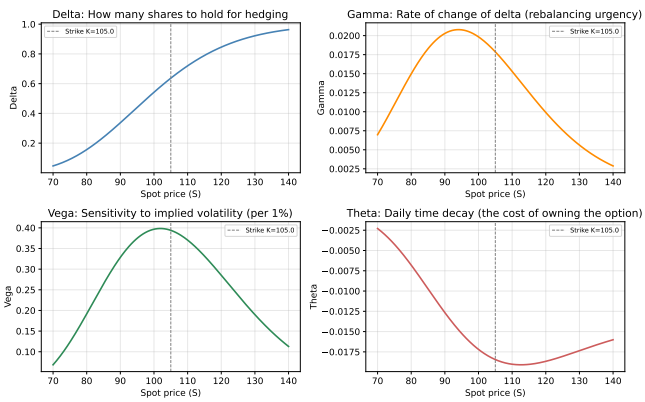

Figure 3: Greeks vs spot price. **Delta** (top-left) rises from 0 (deep OTM) to 1 (deep ITM) in an S-curve, crossing 0.5 near the strike—this is where the outcome is most uncertain. **Gamma** (top-right) peaks at-the-money: this is where delta changes fastest and hedging is most costly. **Vega** (bottom-left) also peaks ATM: at-the-money options are most sensitive to volatility changes. **Theta** (bottom-right) is most negative ATM: the option loses the most value per day here, because that is where uncertainty erodes fastest.

The relationship $\Theta \approx -\frac{1}{2}\Gamma\sigma^2 S^2$ shows that gamma and theta are two sides of the same coin: the option seller collects theta as compensation for bearing gamma risk.

6.2 P&L decomposition

On every trading desk, daily P&L is decomposed by Taylor expansion:

$$\Delta C \approx \underbrace{\Delta \cdot dS}_{\text{delta}} + \underbrace{\tfrac{1}{2}\Gamma \cdot dS^2}_{\text{gamma}} + \underbrace{\mathcal{V} \cdot d\sigma}_{\text{vega}} + \underbrace{\Theta \cdot dt}_{\text{theta}}. \tag{11}$$

Numerical example. Stock moves \$100 → \$103 overnight, volatility unchanged ($d\sigma = 0$). With $\Delta = 0.542$, $\Gamma = 0.020$, $\Theta = -0.017$/day:

- Delta P&L: $0.542 \times 3 = +1.627$
- Gamma P&L: $\frac{1}{2} \times 0.020 \times 9 = +0.089$
- Vega P&L: 0 (vol unchanged)
- Theta P&L: -0.017
- **Explained:** $+1.699$ vs **actual:** $+1.696$

The Taylor expansion explains **100.1%** of the actual BS price change. The residual (higher-order terms) is negligible. In production, all four terms are computed daily for every position.

7 Barrier Options

7.1 Definition

A barrier option is activated or deactivated if the stock crosses a level B during its life. A **down-and-out call** pays $(S_T - K)^+$ only if the stock never touches B from above:

$$\text{Payoff} = (S_T - K)^+ \cdot \mathbf{1}\left\{\min_{0 \le t \le T} S_t > B\right\}. \tag{12}$$

Barrier options are **path-dependent**: the full price trajectory matters, not just S_T. No general closed form exists, making MC the natural pricing tool.

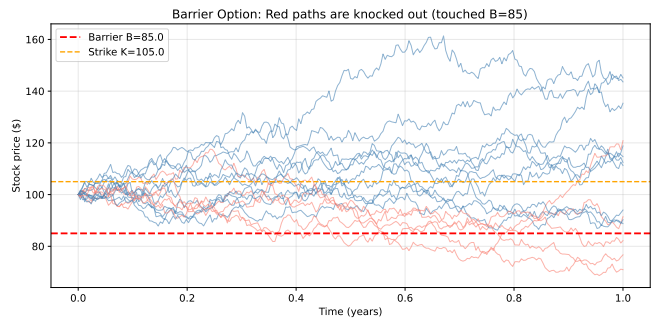

Figure 4: 15 sample paths for a down-and-out call. Blue paths survived (never touched $B = 85$). Red paths were knocked out: some of these end above the strike $K = 105$ and would have been profitable as vanilla calls, but the barrier cancellation zeroes their payoff. This is the risk that makes barrier options cheaper.

7.2 Pricing across barrier levels

Table 3 shows prices for a down-and-out call at various barrier levels. Higher barrier (closer to spot) → more knock-outs → cheaper option. A barrier at $B = 90$ gives a 12.1% discount vs vanilla.

Table 3: Down-and-out call ($K = 105$, daily monitoring).

B	Price	KO rate	Discount
70	8.05	5.2%	−0.4%
80	8.01	20.9%	0.1%
85	7.81	34.6%	2.6%
90	7.05	52.4%	12.1%
95	5.03	73.4%	37.3%
Vanilla	8.02	0%	—

7.3 Discrete monitoring bias and BGK correction

In simulation, the barrier is checked at discrete time steps. Between two grid points, the stock may cross and recover undetected, causing **systematic overpricing** of knock-out options.

Broadie, Glasserman, and Kou [4] showed that the bias is corrected by shifting the barrier:

$$B_{\text{eff}} = B \cdot \exp\left(\beta_1 \, \sigma \sqrt{T/M}\right), \tag{13}$$

where M is the number of monitoring steps and $\beta_1 = -\zeta(1/2)/\sqrt{2\pi} \approx 0.5826$.

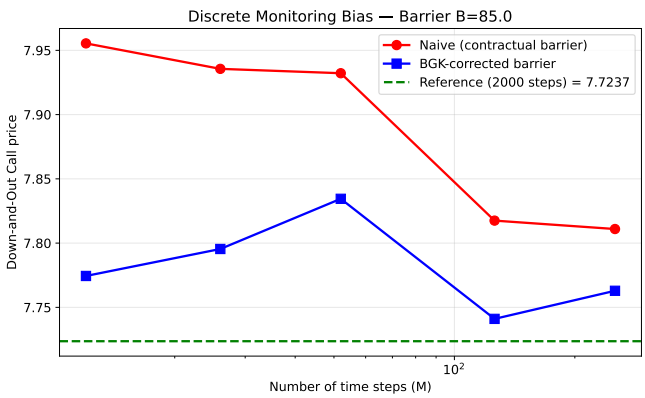

Figure 5: Naive (red) vs BGK-corrected (blue) vs continuous reference (green dashed). The naive estimator systematically overprices the knock-out option because it misses intra-step barrier crossings. The BGK correction shifts the barrier upward to compensate, bringing the price much closer to the continuous-monitoring reference. With monthly monitoring ($M = 12$), the naive bias is \$0.23; the correction reduces it to \$0.05.

8 Model Limitations

8.1 The volatility smile

Black-Scholes assumes a single constant σ for all strikes and maturities. In reality, implied volatility varies by strike—the **volatility smile**.

8.2 Fat tails generate the smile

To prove that the smile is a consequence of non-Gaussian returns, I replace normal draws with Student-t ($\nu = 5$), scaled by $\sqrt{(\nu-2)/\nu}$ to preserve variance. For each of 50 strikes, I price calls by MC, then invert BS (via Brent's method) to extract implied volatility.

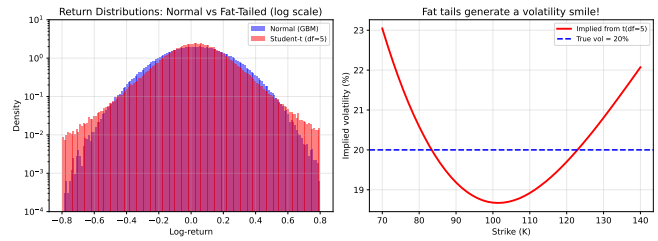

Figure 6: **Left:** Log-scale density of returns. The Student-t distribution (red) has much heavier tails than the normal (blue)—extreme returns occur far more frequently. **Right:** Implied volatility extracted by inverting BS. If BS were correct, the curve would be flat at 20% (blue dashed). Instead, the fat-tailed returns produce a **smile**: implied vol rises in the wings because OTM options are more expensive than BS predicts with flat vol. The kurtosis of the $t(5)$ returns is 8.4 vs 3.0 for normal.

The smile is the market telling us that returns are not normal. Production pricers therefore use local volatility or stochastic volatility models (Heston, SABR), not flat BS.

9 Conclusion

I implemented a Monte Carlo option pricing framework progressing from theory to trading desk practice.

The delta hedging simulation provides the most important conceptual result: the option price is the cost of replication, independent of μ. The fat-tailed simulation provides the most striking empirical finding: the volatility smile emerges mechanically from non-Gaussian returns.

Table 4: Key numerical results.

Metric	Value
MC vs BS error	< 0.03
BGK bias reduction	\$0.23 $\rightarrow$ \$0.05
Taylor P&L explained	100.1%
Daily hedging residual	5.7% of C_0
$t(5)$ kurtosis	8.4 (vs 3.0 normal)

Extensions. Heston stochastic volatility, local volatility calibration via Dupire's formula, Amer-

ican option pricing via Longstaff-Schwartz regression, and multi-asset basket options with correlation.

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